

GPIO Zero Python library

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1. Install GPIO Zero
2. GPIO pin arrangement
3. Pin number
4. Import the GPIO Zero library
5. Reference materials

GPIO Zero is a Python library for beginners, used to control the GPIO interface of the Raspberry Pi;

It provides a simple and easy-to-use interface for controlling peripheral devices such as LEDs, buttons, servos, motors, and various sensors.

1. Install GPIO Zero

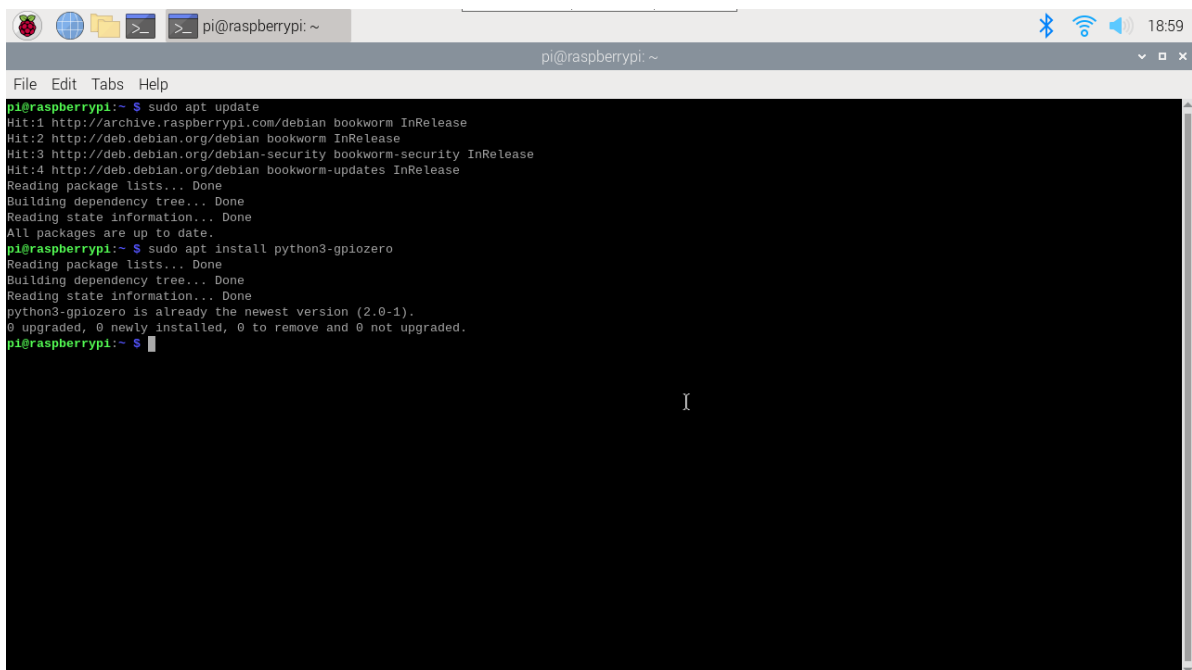
GPIO Zero is installed by default in Raspberry Pi OS desktop images and Raspberry Pi OS Lite images.

- Updated repository list and software

```
sudo apt update
```

- Install GPIO Zero

```
sudo apt install python3-gpiozero
```

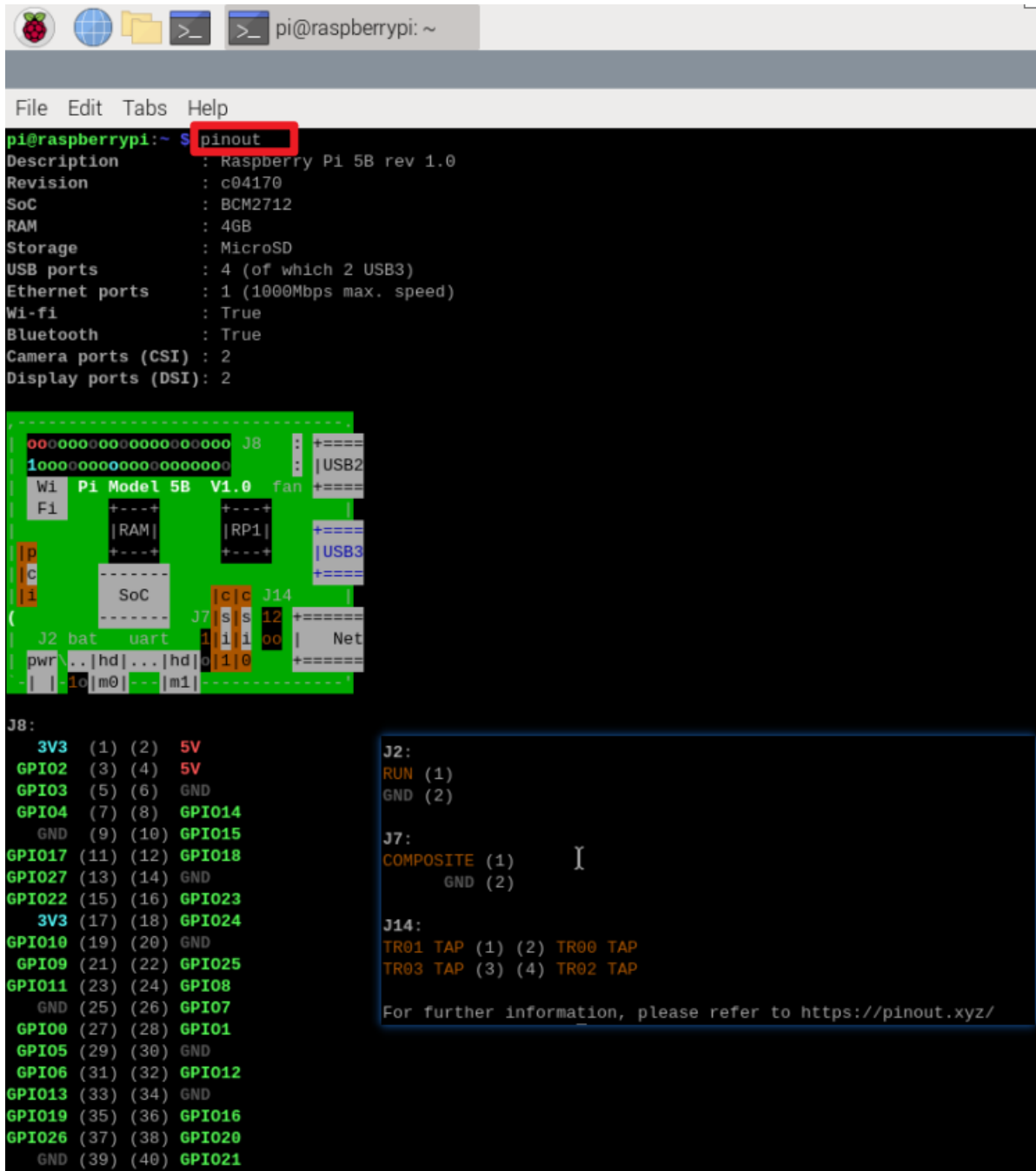
A screenshot of a Raspberry Pi terminal window. The window title is 'pi@raspberrypi: ~'. The terminal shows the output of the command 'sudo apt update', which includes updates for bookworm, bookworm-security, and bookworm-updates. Then, the command 'sudo apt install python3-gpiozero' is executed, showing that the package is already installed at version 2.0-1. The terminal output is as follows:

```
pi@raspberrypi:~$ sudo apt update
Hit:1 http://archive.raspberrypi.com/debian bookworm InRelease
Hit:2 http://deb.debian.org/debian bookworm InRelease
Hit:3 http://deb.debian.org/debian-security bookworm-security InRelease
Hit:4 http://deb.debian.org/debian bookworm-updates InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
pi@raspberrypi:~$ sudo apt install python3-gpiozero
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
python3-gpiozero is already the newest version (2.0-1).
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
pi@raspberrypi:~$
```

2. GPIO pin arrangement

Open the terminal and run the command: pinout

This tool is provided by the GPIO Zero Python library



```
pi@raspberrypi: ~ $ pinout
Description       : Raspberry Pi 5B rev 1.0
Revision          : c04170
SoC               : BCM2712
RAM              : 4GB
Storage           : MicroSD
USB ports         : 4 (of which 2 USB3)
Ethernet ports    : 1 (1000Mbps max. speed)
Wi-fi             : True
Bluetooth         : True
Camera ports (CSI) : 2
Display ports (DSI) : 2

-----
| 00000000000000000000 J8 | : +====
| 10000000000000000000 | : |USB2
| Wi Pi Model 5B V1.0 fan | : +====
| Fi |RAM| |RP1| |         | :
| p | +---+ +---+ +---+ | :
| c | |         | |         | : +====
| i | +---+ +---+ +---+ | : |USB3
|   |         | |         | : +====
|   |         | |         | :
|   |         | |         | :
| J2 bat uart J7 s s 12 | +====
| pwr .. |hd| ... |hd| o | 1 | 0 | Net
| | 10 |m0| --- |m1| |   | +====
|   |         | |         | :

J8:
3V3 (1) (2) 5V
GPIO2 (3) (4) 5V
GPIO3 (5) (6) GND
GPIO4 (7) (8) GPIO14
GND (9) (10) GPIO15
GPIO17 (11) (12) GPIO18
GPIO27 (13) (14) GND
GPIO22 (15) (16) GPIO23
3V3 (17) (18) GPIO24
GPIO10 (19) (20) GND
GPIO9 (21) (22) GPIO25
GPIO11 (23) (24) GPIO8
GND (25) (26) GPIO7
GPIO0 (27) (28) GPIO1
GPIO5 (29) (30) GND
GPIO6 (31) (32) GPIO12
GPIO13 (33) (34) GND
GPIO19 (35) (36) GPIO16
GPIO26 (37) (38) GPIO20
GND (39) (40) GPIO21

J2:
RUN (1)
GND (2)

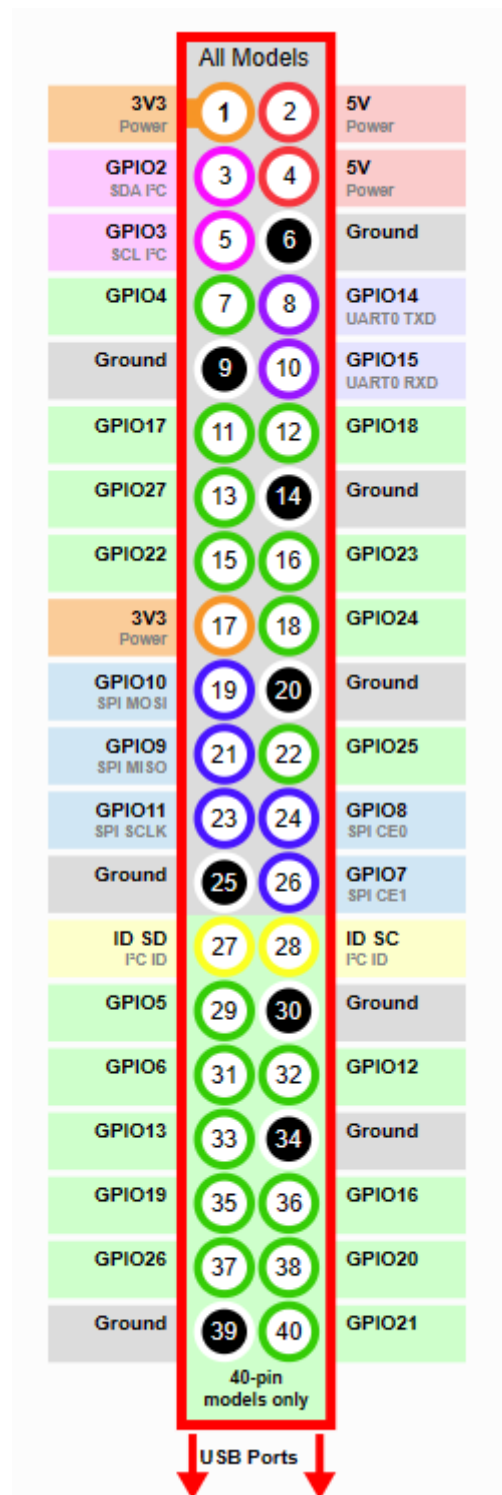
J7:
COMPOSITE (1)
GND (2)

J14:
TR01 TAP (1) (2) TR00 TAP
TR03 TAP (3) (4) TR02 TAP

For further information, please refer to https://pinout.xyz/
```

3. Pin number

The GPIO Zero library uses Broadcom (BCM) pin numbers instead of physical (BOARD) numbers: that is, to control GPIO17, you need to specify 17 instead of 11 for the pin number in the program.



Note: When using the Raspberry Pi GPIO pins, you need to pay attention to the pin connection method between the module and the Raspberry Pi motherboard to prevent damage to the motherboard.

common problem:

LED: When using LED, a current limiting resistor should be added;

Motor: connected through the motor control board/driver board, not directly connected.

4. Import the GPIO Zero library

- Import the entire library

```
import gpiozero
```

- Import a single interface: use the Button interface in the GPIO Zero library

```
from gpiozero import Button
```

5. Reference materials

For more GPIO pin usage, please refer to the tutorial provided on the official website of the GPIO Zero library!

```
official website link: https://gpiozero.readthedocs.io/en/latest/
```